

Proposal: Control-Theoretic and RL-Based Drone Control in 2D Generative Wind Environments

Summary

This project aims to generate wind simulations using generative AI and use these challenging environments to train a drone under difficult disturbance conditions. The drone will first be controlled with a control-theoretic baseline controller, and reinforcement learning will then be integrated to improve adaptation and robustness. In this way, the study investigates whether generative wind scenarios can support more effective and resilient drone control under unseen and complex conditions.



Environment and Generative Wind Modeling

The generative-AI component of this project focuses on building a **2D wind environment for drone control** and learning a model that can generate diverse, controllable wind scenarios for training. The goal is to move beyond a small number of fixed disturbance settings and instead expose controllers to **time-varying, gusty, and chaotic wind conditions** that are more suitable for robust learning.

First, a **physics-inspired 2D environment** will be developed. The drone will be modeled as a rigid body in a top-down 2D space, while the environment will contain a wind vector field that can vary across space and time. The simulator will start with simple wind regimes such as steady flow, gusts, directional shifts, and sinusoidal variations, and will later be extended with **Lorenz-like chaotic forcing** to generate more challenging scenarios. The purpose is not to build a full meteorological model, but a computationally tractable, **episode-based wind simulator** tailored for drone control experiments.

Generative Modeling with CVAE

The generated wind data will then be represented as **wind scenario objects**, rather than as isolated instantaneous values. Each scenario may include temporal wind direction, speed profile, turbulence level, gust events, and spatial asymmetries. These scenario objects will form a synthetic dataset for training a generative model.

For generative modeling, the project will use a **Conditional Variational Autoencoder (CVAE)**. CVAE is a suitable starting point because it learns a compact latent representation of wind scenarios and allows **conditional generation**. This makes it possible to sample scenarios with labels such as *easy / medium / hard*, *steady / gusty / chaotic*, or *low / high turbulence*. As a result, the project will not only generate random wind disturbances, but also **controllable generative wind scenarios** that can be matched to training needs.

A key design choice is to **generate the dataset in-house**. Existing wind simulations[1][2][3] and open tools are either too large-scale, 3D, visualization-oriented, or not directly usable inside a lightweight Python-based RL pipeline. Since this project specifically targets **2D, drone-centered, control-oriented wind environments**, a custom synthetic dataset is more appropriate. This choice is also supported by

recent work[4] showing that large collections of synthetic 2D wind simulations can be successfully used to train learned generative surrogates.

Overall, the expected contribution of this part of the project is a **generative environment pipeline** that produces diverse, physically meaningful, and controllable wind scenarios for downstream reinforcement learning. In this setup, generative AI is used not as a controller itself, but as an **environment distribution designer** for robust drone training.

Drone Control and Reinforcement Learning Relation

The proposed study is primarily grounded in two survey papers. *A Tour of Reinforcement Learning: The View from Continuous Control* [5] provides the main conceptual bridge between reinforcement learning and continuous control, while *Safe Learning in Robotics: From Learning-Based Control to Safe Reinforcement Learning* [6] offers a robotics-oriented perspective on safety, stability, and learning under uncertainty. In addition, several supporting references will be consulted to strengthen the theoretical and applied background of the study: Gullapalli (1992)[7] for the early foundations of RL in control, Lewis et al. (2012)[8] for the relationship between reinforcement learning and feedback control from an optimal and adaptive control perspective, and Hwangbo et al. (2017)[9] for a concrete quadrotor application demonstrating the practical feasibility of RL-based drone control.

First, a control-theoretic baseline controller will be developed for stabilization and trajectory tracking under wind disturbances. Then, an RL-based controller or control augmentation module will be introduced and compared against the classical baseline. Depending on the observed stability and safety requirements, the RL component may be implemented either as a direct continuous-control policy or as a higher-level adaptation layer combined with a safety/control filter. At this stage, the most plausible initial direction is to employ a classical controller together with an RL-based support or adaptation layer, with PPO or SAC as the primary candidate algorithms. Nevertheless, this design choice will remain open to revision as further understanding is developed through the literature review and implementation process.

Evaluation Metrics

Wind Generation

The wind-generation component will be evaluated in terms of realism, physical consistency, diversity, and downstream control utility. Statistical realism will be assessed using **Wasserstein Distance**, **ACF Distance**, **PSD Distance**, and **Ramp-Rate Distance**, while physical plausibility will be measured through **Physical Validity Rate**, **Max Acceleration / Jerk Violation Rate**, and **Torque Plausibility / Wind-to-Torque Consistency**. To evaluate diversity, the study will consider **Pairwise Diversity** and **Scenario Coverage** across different wind regimes such as steady, gusty, and chaotic conditions.

Drone

Controller performance will be evaluated in terms of tracking accuracy, robustness, safety, and control efficiency. The main metrics will include **Position RMSE**, **attitude error**, **settling time**, **recovery time after disturbances**, **crash rate**, **safety-violation rate**, **control effort**, and **control smoothness**. For the RL-based controller, additional metrics such as **sample efficiency**, **generalization to unseen wind scenarios**, and **robustness gain over the classical controller** will also be considered.

Expected Contribution

The main contribution of this project is a control-oriented generative wind environment for drone training and evaluation, together with a comparative study of classical and RL-based control under challenging unseen wind conditions. The study aims to show whether generative wind scenarios can improve controller robustness, generalization, and safety beyond fixed or procedurally randomized disturbances.

References

- [1] <https://www.simscale.com/docs/tutorials/drone-simulation-rotating-zones/>
- [2] <https://raymanng.github.io/3D-Wind-Field/demo/>
- [3] <https://worldwind.arc.nasa.gov/>
- [4] <https://arxiv.org/html/2603.21210v2>
- [5] <https://www.annualreviews.org/content/journals/10.1146/annurev-control-053018-023825>
- [6] <https://www.annualreviews.org/content/journals/10.1146/annurev-control-042920-020211>
- [7] <https://www.proquest.com/docview/304009508?fromopenview=true&pq-origsite=gscholar&sourcetype=Dissertations%20&%20Theses>
- [8] <https://ieeexplore.ieee.org/abstract/document/6315769>
- [9] <https://ieeexplore.ieee.org/abstract/document/7961277>